

Marwan Eid

Software Engineer · Backend & Full-Stack

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WORK EXPERIENCE

Software Engineer · Valeo Jul 2023 – Present

- Designed an event-driven, MQTT-based vehicle communication platform on AWS (EKS, Lambda) for multi-party real-time streaming sessions, placing each session on the least-loaded streaming server and routing device joins through an RDS-backed session registry. Led 2 engineers across system design and delivery.
- Automated CI/CD pipelines with GitHub Actions, Docker and Helm on Kubernetes, cutting deployment time from hours to under 10 minutes with zero-downtime rollouts and automated rollbacks.
- Built an event-driven, closed-loop AI system that detects operational anomalies in CloudWatch logs, generates candidate fixes, and verifies each fix in sandboxed local and demo environments before routing it to engineers for review.
- Cut API response latency from 10 s to 1 s by rewriting cold-start-bound Java Lambda functions in Go.

Junior Software Engineer · Valeo Jul 2022 – Jul 2023

- Incrementally migrated a 75k+ LOC legacy C# desktop application for vehicle lifecycle management (VW, Volvo, HKM) to the web, extracting modular ASP.NET MVC APIs behind a new React frontend. In production for 2+ years, serving hundreds of internal users.
- Built a real-time diagnostics application in C# with an extensible, configuration-driven architecture supporting per-client specifications.

PROJECTS

MilkRun: Real-Time Fleet Tracking (Java, Kafka, PostgreSQL/PostGIS, React) Aug 2026

- Architected a reactive telemetry pipeline with Spring WebFlux and reactor-kafka, streaming GPS events from 50 simulated delivery vans through Kafka to a React/Leaflet dashboard over Server-Sent Events.
- Removed duplicate and out-of-order GPS events with an exact sliding-window deduplicator (8x a Bloom filter's throughput) and per-van reorder buffers, committing Kafka offsets only after idempotent writes.
- Predicted arrivals by map-matching GPS onto planned routes with PostGIS delay zones (~5 s mean error), and federated the live fleet with PostgreSQL history via Apache Calcite.
- Shipped via GitHub Actions CD with automatic rollback; Prometheus and Grafana; 159 tests incl. Testcontainers.

High-Performance Chat System (Ruby on Rails, Go, Redis, MySQL, Elasticsearch) Nov 2025

- Split a chat API into a Go write service (race-free sequence numbers via Redis atomic counters, async persistence through Sidekiq) and a Rails layer for reads and business logic, with Elasticsearch for full-text search.

EDUCATION

B.Sc. in Computer Engineering · The American University in Cairo (AUC) Aug 2017 – Jun 2022

- Minors in Mathematics and Economics · Cum Laude (Honors).
- Recipient of the Atef & Fofa Eltoukhy Family Endowed Scholarship (\$125K), awarded to 1 student every 5 years.
- Represented Egypt at the 2017 Mediterranean Youth Mathematical Championship (bronze medal), qualifying through the Egyptian Mathematical Olympiad.
- Graduation project: **The Right Losses for the Right Gains** (arXiv:2312.10854), a text-to-image architecture that outperformed state-of-the-art baselines, improving FID by 44%.

SKILLS

Languages	Java, Go, C#, Python, TypeScript, JavaScript, C++
Backend & APIs	Spring Boot, WebFlux, ASP.NET (MVC, Core), Node.js, Ruby on Rails, Apache Kafka, MQTT, Protobuf, REST
Cloud & DevOps	AWS (EKS, Lambda, RDS, DynamoDB, SQS), Docker, Kubernetes (Helm), Terraform, GitHub Actions, Jenkins
Data	PostgreSQL / PostGIS, MySQL, MongoDB, Redis (ElasticCache), Elasticsearch, Apache Calcite
Frontend	React, TypeScript, Next.js, Redux, Leaflet
Observability & Testing	Prometheus, Grafana, Micrometer, JUnit, Vitest, JMeter